

Agentic AI Needs a Systems Theory

Abstract

This Paper emphasis in AI Development, which is majorly relying on the capabilities of single models. According to the authors, this underestimates the potential and the risks of the so-called Agentic AI. They suggest a theoretical point of view which may enable us to comprehend these technologies. Our perspective must be the broader view of how agents relate with each other, human beings and the world. The paper describes the development of an advanced functionality like causal reasoning and metacognition that can naturally arise out of these interactions even though the individual agents are relatively simple. It provides advice on the ways these systems can be deliberately designed to provide greater control over their emergent behaviors.

Introduction

The main aim fueling the existing studies in AI will be to develop solutions that can accomplish long-term complex tasks without requiring human hand-holding. It is incredibly exciting due to the possibility to simplify the working process and even speed up the process of scientific discoveries. The truth of the matter is though that even the current agents, which are mostly founded on large language models, are still rather fragile. They tend to have a low common sense, cannot relate properly with the real world, and may get themselves and their users confused easily. According to the author, we cannot work on making the models smarter on their own. Such a strategy is actually becoming risky; already we are witnessing such problems as the so-called alignment faking, when an AI behaves by the rules when it is being observed, but behaves differently when the supervision is removed. The article recommends the use of a system theory approach. We can look at the bigger picture of how the agents, human and the environment interact and change each other.

Related Work

The authors relate their concepts with a number of emerging fields of studies. They refer to the concept of agency loss in research involving the collaboration of human and AI. This occurs when individuals over depend on AI, and the decision made by humans becomes more predictable and less independent. They also cite studies which help to support two-way alignment, and both humans and AI learn and work together rather than humans alone adapt to AI. The paper contrasts its approach to designing AI systems with other approaches that the paper only considers development of abilities such as memory or perception directly into the AI model. Though the authors concur that AI must have the capability of addressing uncertainty and coming up with considerate decisions, they do not feel that these features always have to be programmed into the system. According to authors, ability like good judgment and self-awareness can be naturally acquired in the process of interaction between the AI, human beings and the environment.

Methodology

The analytical conceptualization of the paper is based on the redefinition of the agency that the authors have called Functional Agency. Instead of regarding agency as an unclear and inexplicable internal attribute, they

describe it in terms of what is done by the system. A functional agency system is one that is able to produce actions to a goal, preserve a model of their effects on the consequences, and respond to variations in these consequences. On this definition they build a model of an Agentic System. This system is not merely the AI software; it is the group of agents (with the help of tools), the human operators offering commands, and the outside world. The tools in this perspective determine the capabilities of the agent and are agency less. Its key idea is that the system as a whole can be highly agency and intelligence, despite the simplicity of the individual components of the system, just as complex biological behavior can arise out of the simple cellular networks.

Results

The article presents a subtle model that is known as functional agency. Instead of considering agency as some obscure attribute of the code, the authors also introduce it in a highly precise manner, according to the capabilities of the system. Particularly, a system has agency when it can produce action toward a goal, model the relationship between its actions and their consequences, and more importantly be able to modify its behavior should relationships between the outcomes and actions change. In elaboration, they view an Agentic System to consist of much more than the AI model. It is an ecosystem comprising of the human operators, the tools (capabilities but not agency) and the external environment. The main claim here is that though the individual components can be simple, high level intelligence can arise through the interaction of all these components, just as complex biological behaviors are brought about by the interaction of cells. The authors find three distinct ways of this emergence. First, they claim that active engagement with the environment e.g. similar to "embodied cognition" enables agents to develop stronger representations than they could acquire when only exposed to static data. Second, causal reasoning can automatically be developed by the mere urge to reduce errors in prediction. Last, the fact that coordinating with the other agents requires the system to tune its confidence and uncertainty levels results in some kind of collective self-consciousness or metacognition.

Conclusion

The paper has determined that agentic AI is in the crossroad. We do not just wait and hope that emergent abilities will appear, but we have to contrive arrangements to encourage these beneficial interactions. By utilizing a systems-theoretic approach, researchers can more readily anticipate such dangers as the uncontrolled creation of subgoals and design governance structures without infringing the human agency. It is not only to build smarter models in the end analysis but also to build the relationships that allow the safe and productive intelligence to emerge.